

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

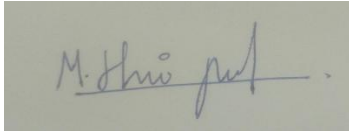
Total Marks: 25

Code of Conduct

I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1.A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Cloud Auto-Scaling

Cloud auto-scaling automatically increases or decreases the number of cloud resources based on application demand. During seasonal sales, such as Black Friday or Diwali, website traffic can increase suddenly. Auto-scaling detects this increase and launches additional virtual machines or instances to handle the extra workload. When traffic decreases after the sale, the extra resources are removed automatically. This ensures efficient resource utilization and reduces unnecessary costs because the company pays only for the resources it uses.

Load Balancing

Load balancing distributes incoming user requests evenly across multiple servers. Instead of sending all requests to one server, it directs traffic to the available and least busy servers. If one server becomes overloaded or fails, the load balancer automatically redirects requests to another healthy server. This improves system reliability, prevents server overload, and maintains continuous service.

Reducing Downtime and Improving User Experience

The combination of auto-scaling and load balancing helps reduce downtime by ensuring that enough servers are available during periods of high demand. Users experience faster page loading, smooth browsing, successful online payments, and uninterrupted checkout processes. This improves customer satisfaction and prevents revenue loss caused by website failures.

Role of Monitoring Tools

Monitoring tools continuously monitor system performance by tracking CPU utilization, memory usage, network traffic, response time, and server health. When resource usage exceeds predefined limits, monitoring tools automatically trigger auto-scaling to add more servers. They also generate alerts for administrators if any server experiences performance or security issues. This ensures proactive management and high application availability.

Real-World E-Commerce Scenario

During **Amazon Prime Day**, millions of customers visit the Amazon website at the same time. Amazon uses cloud auto-scaling to automatically launch additional servers as traffic increases. Load balancers distribute customer requests evenly across these servers, preventing overload. Monitoring tools continuously observe system performance and automatically adjust resources whenever required. This enables customers to browse products, make payments, and place orders without delays or website crashes.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case

Durability and Data Redundancy

Cloud storage ensures durability by storing multiple copies of backup data across different storage devices and geographically separated data centers. If one storage device or an entire data center fails due to hardware failure, power outage, or natural disaster, the data can still be accessed from another location. This significantly reduces the risk of permanent data loss.

Data redundancy means maintaining duplicate copies of data in multiple locations. The cloud provider automatically replicates backup files so that even if one copy becomes corrupted or unavailable, another copy remains available. This ensures continuous availability of backup data and improves business continuity.

Encryption and Access Control

Cloud providers protect sensitive backup data using encryption and access control.

Encryption converts backup data into an unreadable format while it is stored (encryption at rest) and while it is transmitted over the network (encryption in transit). Only users with the correct encryption keys can decrypt and access the original data.

Access control ensures that only authorized users can access backup files. Cloud providers use Identity and Access Management (IAM), Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), and user authentication to prevent unauthorized access and protect confidential business information.

Cost Advantages Compared to Traditional Storage

Cloud storage is more cost-effective than traditional storage systems because organizations do not need to purchase expensive storage hardware, maintain servers, or invest in additional data center infrastructure. Cloud providers follow a pay-as-you-go pricing model, where businesses pay only for the storage they actually use.

Cloud storage is also easily scalable. As backup requirements increase, additional storage can be added instantly without purchasing new hardware. This reduces maintenance costs, electricity costs, and administrative overhead while providing greater flexibility.

Disaster Recovery Use Case

Consider a company whose on-premises data center is damaged due to a flood or fire. Since all critical business data has been regularly backed up to cloud storage, the company can quickly restore its applications, databases, and files from the cloud. Employees can resume business operations with minimal

downtime and without significant data loss. This enables faster disaster recovery and ensures business continuity.

3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Content Delivery Networks (CDNs)

A Content Delivery Network (CDN) is a distributed network of servers located in different geographic regions. It stores cached copies of frequently accessed content such as images, videos, application files, and web pages. Instead of retrieving content from the main server every time, users receive it from the nearest CDN server.

This reduces the distance that data must travel over the internet, resulting in faster content delivery, lower network congestion, and improved application performance. CDNs are especially useful for applications with users spread across multiple countries.

Reducing Latency with CDNs

Latency is the time taken for data to travel between the server and the user. When users access content from a nearby CDN server instead of a distant central server, the response time is significantly reduced.

Lower latency provides several benefits:

- Faster application loading.
- Reduced buffering during video playback.
- Quicker file downloads.
- Faster response to user requests.
- Improved overall application performance.

This ensures a smooth experience even when many users access the application simultaneously.

Role of Edge Servers

Edge servers are CDN servers located close to end users in different regions. They store cached copies of frequently requested content and deliver it directly to nearby users.

By serving content from the closest location, edge servers:

- Reduce response time.
- Minimize buffering and delays.
- Improve application speed.
- Reduce the workload on the central server.

- Provide a consistent user experience regardless of geographical location.

This is especially important for applications that require real-time communication or continuous data streaming.

Global Availability through Cloud Providers

Cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) maintain multiple data centers across different regions and availability zones worldwide. Applications can be deployed in several regions simultaneously, ensuring high availability, fault tolerance, and disaster recovery.

If one data center experiences a failure, traffic is automatically redirected to another available region. This ensures that the application remains accessible with minimal downtime for users around the world.

Real-Time Streaming or Gaming Example

Netflix uses CDNs and edge servers to deliver movies and TV shows from servers located close to users. This reduces buffering, improves video quality, and provides smooth streaming even during peak viewing hours.

Similarly, online multiplayer games such as PUBG or Fortnite use edge servers to reduce latency between players and game servers. This enables faster response times, smoother gameplay, and a better gaming experience for users across different countries.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Regulatory Compliance and Auditing

Financial institutions handle highly sensitive customer information and must comply with regulations such as PCI DSS, ISO 27001, GDPR, and local banking regulations. Cloud providers support regulatory compliance by offering certified cloud infrastructure, built-in security controls, regular security assessments, and compliance reports.

Cloud providers also maintain detailed audit logs that record user activities, system changes, login attempts, and data access. These logs help organizations verify compliance, investigate security incidents, and provide evidence during internal and external audits. Regular auditing ensures that security policies are followed and regulatory requirements are met.

Role of Logging, Monitoring, and Encryption

Logging records all user and system activities, including logins, file access, configuration changes, and financial transactions. These records help track user actions and detect unauthorized access.

Monitoring continuously observes system performance, network traffic, and user behavior. It identifies unusual activities such as unauthorized login attempts or suspicious transactions and immediately alerts administrators, enabling quick response to security threats.

Encryption protects sensitive financial data both at rest (stored in cloud databases) and in transit (while being transferred over the internet). Even if data is intercepted by attackers, it cannot be read without the correct encryption keys, ensuring confidentiality and integrity.

Ensuring Data Sovereignty

Data sovereignty means that an organization's data must be stored and processed according to the laws of the country where it is collected. To meet these requirements, the financial institution should select cloud regions located within the required country or geographical area.

Cloud providers allow customers to choose the storage location of their data, helping organizations comply with national regulations and ensuring that customer information does not leave the approved region.

Example of Sensitive Financial Transactions

Consider an online banking system where customers transfer money using a mobile banking application. During the transaction, customer details and payment information are encrypted before transmission and remain encrypted while stored in the cloud. Every transaction is logged for auditing, while monitoring

tools continuously check for suspicious activities such as unusual login attempts or fraudulent transfers. The financial data is stored in cloud

5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containers vs Virtual Machines

Containers and Virtual Machines (VMs) are both used to deploy applications, but they differ in architecture and resource usage. Containers share the host operating system kernel and package only the application and its dependencies. In contrast, Virtual Machines include a complete guest operating system along with the application.

Because containers do not require a separate operating system, they are lightweight, start within seconds, and consume less CPU, memory, and storage. This allows more containers to run on the same hardware compared to virtual machines. Virtual machines require more resources, have longer boot times, and are better suited for running different operating systems on the same physical server.

Role of Kubernetes in Deployment and Scaling

Kubernetes is a container orchestration platform that automates the deployment, management, and scaling of containerized applications. It simplifies application deployment by scheduling containers on available servers and managing them automatically.

Kubernetes continuously monitors the health of containers and restarts them if they fail. It also supports automatic scaling, where additional container instances are created during high traffic and removed when demand decreases. Features such as load balancing, rolling updates, and automatic recovery help maintain high availability and minimize downtime without manual intervention.

Portability Across Different Cloud Environments

One of the major advantages of containers is portability. Since containers package the application along with all required libraries and dependencies, the same container image can run consistently on different cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), or even on on-premises servers.

This eliminates compatibility issues between different environments, simplifies application migration, and reduces vendor lock-in. Developers can build the application once and deploy it anywhere without making changes to the code.

Example of a Microservices-Based Application

An e-commerce application is a common example of a microservices-based architecture. The application can be divided into independent services such as:

- **User Authentication Service** – Manages user login and registration.
- **Product Catalog Service** – Displays product information.

- **Shopping Cart Service** – Handles items added to the cart.
- **Payment Service** – Processes online payments.
- **Order Management Service** – Manages order placement and tracking.

Each microservice runs inside its own container and is managed by Kubernetes. If customer traffic increases during a sale, Kubernetes can automatically scale only the Payment or Product Catalog services without affecting the other services. This improves scalability, reliability, and application performance.